Author (A) Paper (P ) Subject (S )

PAP

PSP

(a) A heterogeneous graph and predefined meta- paths

Node Attention

Node Attention

Semantic Attention

0.2 → 32 bit

0.2

0.1

0.7

(b) Hierarchical aggregating paradigm of HAN

Shared Conv

Attention

Integrate

Fire

Reset

SNN

1

0

1 bit

(c) Aggregating paradigm of SpikingHAN